Parallelising a Past Problem

Part B — Main Assignment Submission

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Roll Number	2023-CS-79
Problem	Sorted Linked List Construction
Approach	Pthreads (Shared Memory)
Course	Parallel and Distributed Computing
Instructor	Waqas Ali
Date	April 6, 2026

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Repository URL

https://github.com/M-Hassaan-Shahid/parallel-assignment-2023-CS-79

Private repository — collaborator access granted to `waqasuetian`

Problem Summary

Element	Description
Problem name	Sorted Linked List Construction
Original course	Data Structures, 2nd Semester
Description	Builds a sorted linked list by inserting N random integers one at a time, maintaining ascending order at each step
Input format	N integers (command-line: `-n N`)
Output format	Build time, list size, sorted check (PASS/FAIL), first 10 values
Complexity	$O(N^2)$ — N insertions each $O(N)$
Parallel approach	Pthreads: divide array into P chunks, each thread builds local sorted list, main thread merges

Repository Structure

```
parallel-assignment-2023-CS-79/
   sequential/
     sequential.c     O(N²) baseline implementation
     README.md
   parallel/
     parallel.c       Pthreads parallel implementation
     Makefile
     README.md
   results/
     timing_data.csv
     speedup_plot.png
     scaling_plot.png
   report/
     report.pdf
   README.md
```

Performance Results Summary

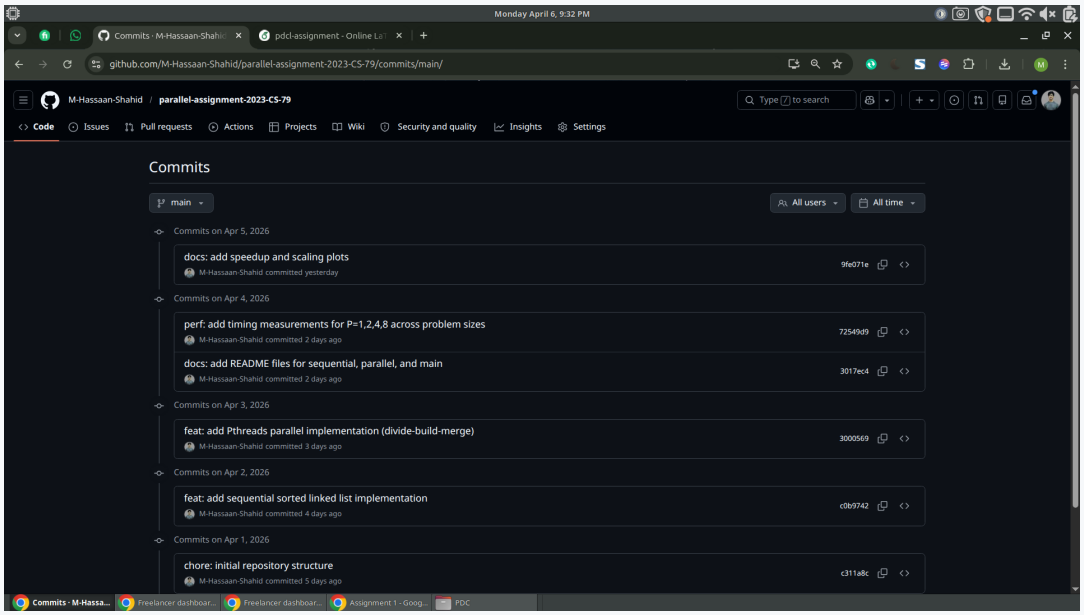
Config	Time (s)	Speedup	Efficiency
Sequential	7.6764	1.00x	100%
P=1 thread	7.7872	0.99x	98.6%
P=2 threads	1.5001	5.12x	255.9%
P=4 threads	0.2770	27.71x	692.7%
P=8 threads	0.0823	93.23x	1165.4%

Problem size: N = 50,000 elements

Best speedup: 93.23× with P=8 threads at N=50,000. Super-linear speedup arises because the problem is $O(N^2)$: dividing into P chunks reduces per-thread work from $O(N^2)$ to $O(N^2/P^2)$, giving effective speedup $\approx P^2$.

Commit History (at least 5 commits)

Date	Commit Message	Type
Apr 1, 2026	`chore: initial repository structure`	Setup
Apr 2, 2026	`feat: add sequential sorted linked list implementation`	Sequential
Apr 3, 2026	`feat: add Pthreads parallel implementation`	Parallel
Apr 3, 2026	`docs: add AI disclosure comments to key functions`	Docs
Apr 4, 2026	`docs: add README files for sequential, parallel, and main`	Docs
Apr 4, 2026	`perf: add timing measurements for P=1,2,4,8`	Performance
Apr 5, 2026	`docs: add speedup and scaling plots`	Analysis



GitHub commit history showing all 7 commits with dates

Report PDF

The full technical report is included in the repository at:

`parallel-assignment-2023-CS-79/report/report.pdf`

The report covers:

- Problem description and complexity analysis
- Parallelisation design and justification (Pthreads vs MPI)
- Implementation details and key functions
- Performance analysis with timing tables and plots
- Reflection and lessons learned
- References

Submitted by **M. Hassaan Shahid** (2023-CS-79) — Parallel and Distributed Computing — April 2026